PROGRESS PRESENTATION

DORMATRIX

Centralized CLI Dormitory Management System

TEAM 1

TEAM MEMBERS

Ayman Binta Altaf Nondiny - 230042141

Khadiza Sultana - 230042135

Procheta Silvie - 230042114

Sayma Tasnim - 230042139

PROJECT **OVERVIEW**

Centralized CLI system for dormitory management

Key Objectives:

- Efficiently manage student complaints
- Organize facility bookings
- Facilitate departmental store operations
- Reduce manual work and improve transparency

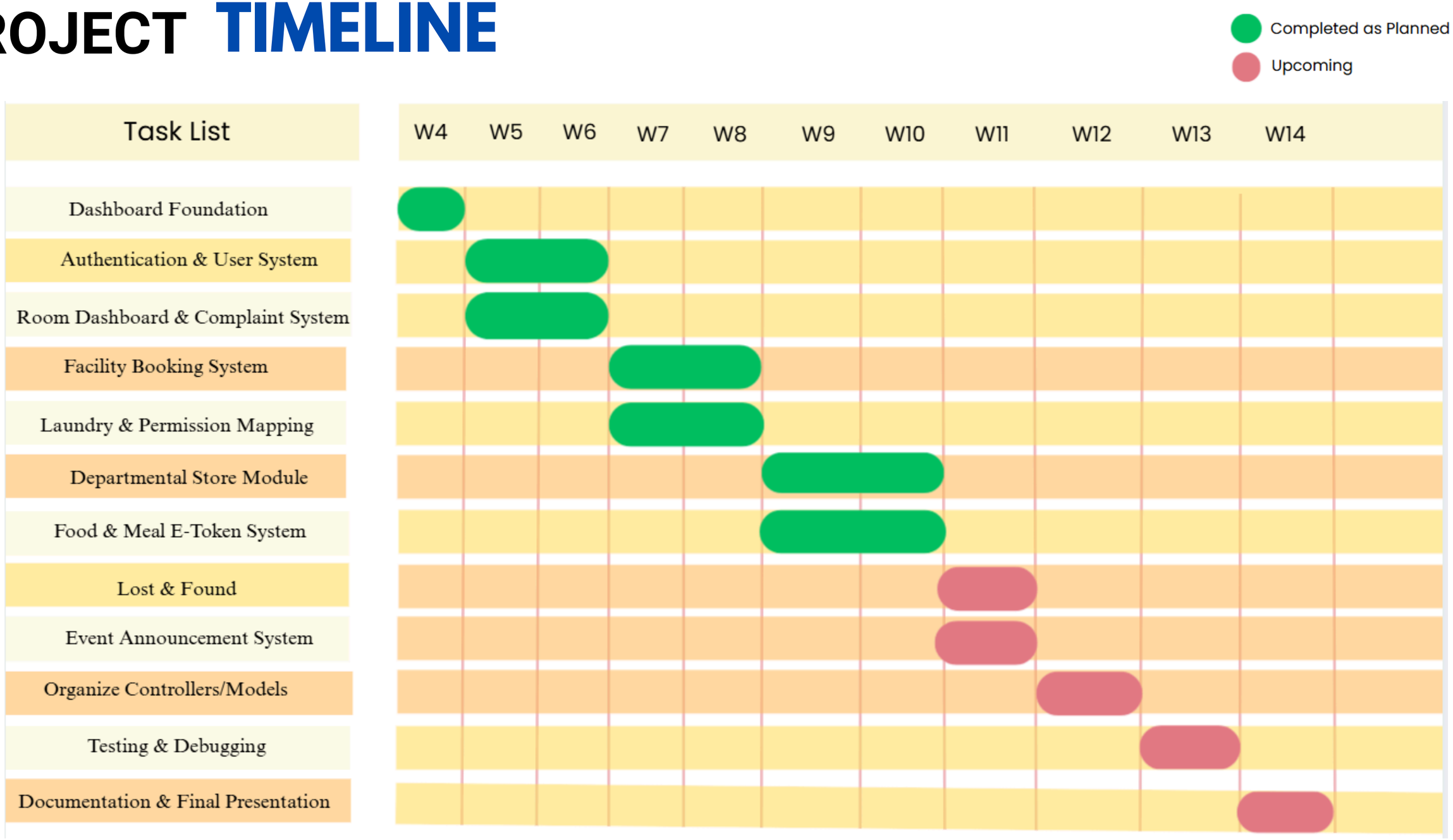
CORE STRUCTURE & AUTHENTICATION

- Modular CLI-based system architecture established
- Centralized navigation and menu flow implemented
- Secure user authentication mechanism integrated
- Role-based access control enforced for different user types
- Controlled access to system features based on assigned role

TOOLS & TECHNOLOGIES

- **Language :** Java
- **File Based Storage :** text based file (.txt)
- **I/O:** Java FileReader / FileWriter / BufferedReader
- **Architecture:** Modular OOP
- **Version Control:** Github
- **SDLC Model:** Iterative + Incremental
- **Slot Booking Algorithm** : First Fit Allocation
- **Development Environment:** IntelliJ IDEA

PROJECT TIMELINE



FUNCTIONALITIES IMPLEMENTED SO FAR

1. Complaint Management
2. Slot Booking System
3. Departmental Store Module

COMPLAINT MANAGEMENT

- Student complaint submission
- Attendant and maintenance staff can view complaints
- Initial framework for complaint categorization and status updates

COMPLAINT MANAGEMENT

```
Select Complaint Category:
1. ELECTRICITY
2. PLUMBING
3. INTERNET
4. CLEANING
5. SECURITY
6. OTHER
Enter choice: 1
Enter complaint description: not available

Complaint Filed Successfully!
Complaint ID : C-3
Status       : ASSIGNED
Assigned To  : MW01



------------------ COMPLAINTS (STUDENT) ------------------
1. File a Complaint
2. View My Complaints
3. Track Complaint by ID
0. Back
Enter choice: 2

------------------ LIST ------------------
ID: C-3 | Student: 230042114 | Room: 302 | Cat: ELECTRICITY | Status: ASS
```

DEMONSTRATION I

2 SLOT BOOKING MANAGEMENT

- Facility booking for:
  - Study rooms
  - Refrigerator slots
  - Laundry slots
- Logic for preventing conflicts in bookings

2 SLOT BOOKING MANAGEMENT

```
|                        STUDENT DASHBOARD                          |
-------------------------------------------------------------------
  Welcome, Sayma Tasnim
-------------------------------------------------------------------
| 1. View Room Info                                                |
| 2. Facility Booking                                              |
| 3. Meal Token Purchase                                           |
| 4. Store Purchases & Dues                                        |
| 5. Lost & Found                                                  |
| 6. Submit Complaint                                              |
| 7. View Announcements                                            |
| 0. Logout                                                        |
-------------------------------------------------------------------
Enter your choice: 2


-------------------------------------------------------------------
|                      Facility Booking System                     |
-------------------------------------------------------------------
| 1.Book Study Room Seat                                           |
| 2.Check-in to Study Room                                         |
| 3.Book Fridge Slot                                               |
| 4.Schedule Laundry Machine                                       |
| 0. Logout                                                        |
-------------------------------------------------------------------
Enter your choice: 1
Seat reserved! You have 30 seconds to Check-in (Click Option 2).
```

DEMONSTRATION II

3 DEPARTMENTAL STORE MODULE

- Inventory listing
- Recording student purchases
- Basic dues ledger

3 DEPARTMENTAL STORE MODULE

```
| 1. View Room Info
| 2. Facility Booking
| 3. Meal Token Purchase
| 4. Store Purchases & Dues
| 5. Lost & Found
| 6. Submit Complaint
| 7. View Announcements
| 0. Logout
-------------------------------------------------------------------------
Enter your choice: 3
-----------------------------------------------
|              Meal Token System              |
-----------------------------------------------
| 1. Buy Meal Token                           |
| 2. Validate Meal Token                      |
| 0. Back                                     |
-----------------------------------------------
Enter choice: 1
Token purchased successfully!
Token ID: MT-ZY4L
```

DEMONSTRATION III

FUTURE ENHANCEMENTS &

EXPANSION PLANS

- Lost & Found Module
- Announcement System
- Facility Enhancements
- System Improvements (as required)

THANK YOU